Fashion Recommendation via Graph Neural Networks, Hybrid Collaborative Models and Web-Crawled Multimodal Data

Nguyen-Phuc Mac Qui-Hung Kieu Duy-Liem Nguyen Tri-Duc Tran

Faculty of Information Science and Engineering,
University of Information Technology - VNUHCM
{22521123,22520505,22520752,22520276}@gm.uit.edu.vn

Abstract

This project addresses the fashion recommendation problem using a large-scale multimodal dataset collected from Uniqlo Vietnam. By comprehensively evaluating methods ranging from traditional approaches to advanced Graph Neural Network (GNN) architectures (LightGCN, NGCF, MMGCN, LightGCL), the study focuses on leveraging the effectiveness of contrastive learning and feature fusion. Experimental results demonstrate the superiority of multimodal graph-based models and highlight the crucial role of textual features and multilingual embeddings in improving recommendation accuracy within the Vietnamese market.

Keywords: Recommender Systems, Fashion Recommendation, Graph Neural Networks, Multimodal Learning.

1 Introduction

The rapid growth of e-commerce, particularly in the fashion domain, has resulted in substantial information overload, creating significant challenges for consumers in searching for and selecting suitable products. In this context, *Recommender Systems* play a pivotal role in personalizing user experiences and supporting decision-making. To address this problem, the research community has developed a diverse set of methodological approaches, ranging from foundational methods such as *Content-Based Filtering*, which leverages product attributes, to *Collaborative Filtering*, which exploits community interaction patterns. Each approach embodies a distinct underlying philosophy, optimizing predictions based on either data-driven product characteristics or the collective intelligence of the user community.

Beyond traditional approaches, the emergence of *Graph Neural Networks* (GNN) and *Multimodal Deep Learning* has opened new directions for modeling complex interaction data. Advanced architectures such as NGCF, LightGCN, MMGCN, and LightGCL enable representing user–item relationships as graphs while integrating diverse information sources (e.g., images, text) to improve predictive accuracy. In this project, we conduct a comprehensive empirical evaluation on a real-world dataset from Uniqlo to compare the effectiveness of methods ranging from traditional to modern. The objective is to analyze operational characteristics and identify the unique strengths of each strategy within the context of fashion recommendation, thereby providing a reference for selecting appropriate models for practical application scenarios.

2 Related works

The combination of *Graph Neural Networks – GNN* and *Multimodal Deep Learning* has driven significant progress in recommender systems, particularly within the fashion domain. This project builds upon and further develops three major academic directions, as detailed below.

2.1 GNN-based Collaborative Filtering

Modeling interactions as a bipartite graph has demonstrated superior effectiveness compared to matrix factorization. Wang et al. (2019) introduced *Neural Graph Collaborative Filtering* (NGCF), which leverages *high-order connectivity* through a *message passing* mechanism; however, this architecture incurs substantial computational overhead. To address this limitation, He et al. (2020) proposed *LightGCN*, an efficient *baseline* model that simplifies GCN architectures by removing nonlinear transformations and retaining only linear *neighborhood aggregation*, thereby producing smoother representations with lower computational complexity.

2.2 Multimodal Recommendation in Fashion

The inherent characteristics of fashion require deep understanding of both visual and seman-

tic cues. While *Multimodal Attention Network* (Chen et al., 2019) and *FashionBERT* (Gao et al., 2020) emphasize the fusion of multi-source data (images, text), Wei et al. (2019) advanced this line of research from a graph-based perspective through the *Multi-modal Graph Convolution Network* (MMGCN). Rather than simply concatenating features, MMGCN constructs separate modality-specific subgraphs, enabling the model to capture fine-grained user preferences across visual or semantic dimensions—an especially suitable approach for complex multimodal datasets.

2.3 Enhancing Robustness with Contrastive Learning

To address issues of data sparsity and noise, Cai et al. (2023) proposed *LightGCL*, which integrates contrastive learning into recommender systems. Unlike common random augmentations that risk disrupting structural information, LightGCL employs *Singular Value Decomposition – SVD* to generate globally coherent contrastive views. This strategy enables effective graph reconstruction, helping the model learn robust and stable representations even under limited-data conditions.

3 Data Collection and Description

3.1 Data Collection Procedure

To ensure both accuracy and diversity of the experimental dataset, this project implements an automated data collection pipeline from the Uniqlo Vietnam website, combining Selenium to handle dynamically rendered content and BeautifulSoup4 for HTML parsing tasks. The system is designed with a hierarchical structure, sequentially traversing major product categories and simulating user behavior (scrolling, loading asynchronous content) via `WebDriverWait` to maintain stability in the presence of network latency.

The data extraction process is performed in two stages: (1) the *Category Layer*, which gathers basic metadata (name, price, product links); and (2) the *Detail Layer*, which accesses individual product pages to retrieve technical descriptions, material compositions, high-resolution images, and user review histories. After collection, the raw data undergoes exception handling and formatting normalization to remove noise and ensure consistency before entering the preprocessing and model training phases.

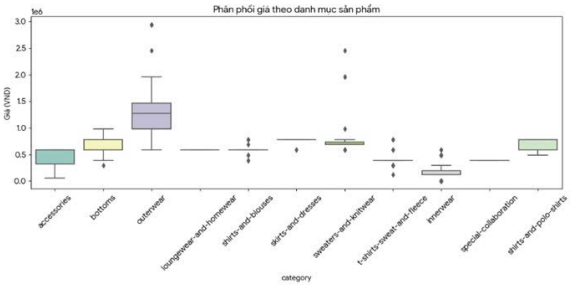

Figure 1: Price distribution by product category.

3.2 Dataset Construction

Following the data collection and preprocessing stages, the dataset is organized into multiple separate files to accommodate various analytical and modeling objectives. A quantitative overview of the dataset is presented in Table 1.

3.2.1 Product Information File

This file serves as the master reference dataset, storing product identifiers and technical specifications. A detailed list of attributes is presented in Appendix B.

3.2.2 Customer Feedback File

This file constitutes the primary data source for sentiment analysis and behavior prediction. During preprocessing, all records lacking user identifiers (Reviewer = "NaN") were removed entirely. Filtering out anonymous accounts without authenticated interaction history is essential to prevent noise and ensure accuracy in personalized recommendation models. The cleaned data structure is described in Appendix B.

3.2.3 Data Characterization via Visualization

To gain deeper insights into the structure and distribution of the dataset, a statistical analysis was conducted using four key visualizations that capture fundamental aspects related to products, customers, and feedback quality.

Regarding product and pricing characteristics (Figures 1, 2), the data clearly reflects the brand's focus on essential product lines, with a dominance of the *Innerwear* and *T-shirts* categories. The price distribution reveals a distinct stratification: most products fall within an accessible price range (160,000–400,000 VND), while *Outerwear* forms a separate group with the highest average price (above 1.3 million VND).

For customer characteristics (Figure 3), demographic distributions indicate a strong concentra-

Dataset	Quantity	Number of Attributes	Description
Product	352	13	Identification, technical, and commercial information of products.
Reviewer	4457	13	Customer review information, comments, and anthropometric attributes.

Table 1: Quantitative statistics of the dataset.

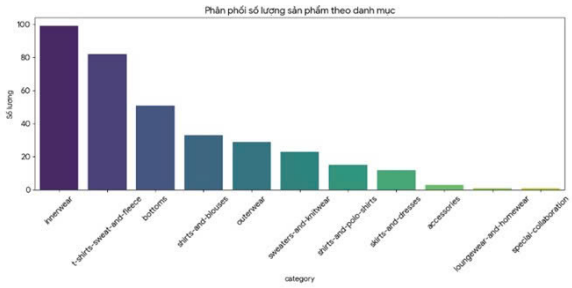

Figure 2: Distribution of product counts across categories.

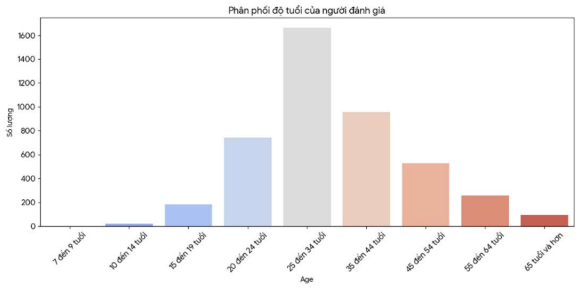

Figure 3: Age distribution of reviewers.

tion of users aged 25 to 44, accounting for more than 60

Regarding feedback quality (Figure 4), the data exhibits a pronounced positive skew, with 5-star ratings comprising over 92

4 Methodology

4.1 Problem Definition and Notation

The problem considered in this project is the Top-K recommendation task in recommender systems, where the input consists of implicit user–item interactions. Let $U = u_1, u_2, \ldots, u_{|U|}$ denote the set of users and $I = i_1, i_2, \ldots, i_{|I|}$ the set of items. The interaction set $R = (u, i)$ represents all pairs in which a user $u \in U$ has interacted with an item $i \in I$.

The objective of the model is to learn a scoring function $y_{ui} = f(u, i)$ such that, for each user u, items previously interacted with receive higher scores than non-interacted ones. This allows the

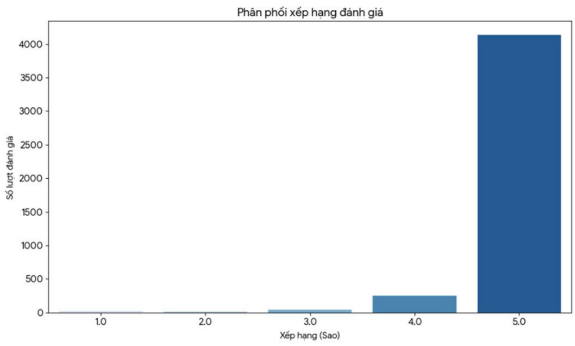

Figure 4: Distribution of review ratings.

model to rank items and select the Top-K highest-scoring items for recommendation.

4.2 Construction of the Interaction Graph

From the interaction data, a bipartite graph $G = (U \cup I, E)$ is constructed, where each edge $(u, i) \in E$ corresponds to an interaction between user $u \in U$ and item $i \in I$. This graph enables direct modeling of user–item relationships and serves as the foundation for information propagation in graph-based recommender models.

During embedding propagation, the adjacency matrix is symmetrically normalized as follows:

$$\hat{A} = D^{-\frac{1}{2}} A D^{-\frac{1}{2}}, \tag{1}$$

where A denotes the adjacency matrix and D the degree matrix. This normalization ensures numerical stability during training and prevents high-degree nodes from dominating the representation learning process.

4.3 Embedding Initialization

Each user and each item is represented by an initial embedding vector in the latent space, denoted by $\mathbf{e}_u^{(0)}$ and $\mathbf{e}_i^{(0)}$, respectively, with $\mathbf{e}_u^{(0)}, \mathbf{e}_i^{(0)} \in \mathbb{R}^d$. These embeddings are randomly initialized and iteratively updated during training through graph-based information propagation and loss-function optimization.

4.4 Content-Based Filtering

This approach exploits intrinsic product features rather than interaction data, using text embeddings and image embeddings. The combined representation is a linear fusion of these two modalities. A *user profile* is constructed by averaging the embeddings of interacted items, and recommendation scores are computed via cosine similarity. While this method effectively addresses the item *cold-start* problem, it suffers from limited *exploration* capability.

4.5 User-Based Collaborative Filtering

This method leverages the interaction matrix to identify groups of users with similar behavioral patterns based on cosine similarity. Recommendation scores are computed by aggregating the interaction histories of the k most similar users. This approach harnesses the *wisdom of the crowd* without requiring item features; however, it faces scalability challenges and struggles with the *cold-start* problem for new users.

4.6 LightGCN

LightGCN is selected as the baseline model due to its simplified architecture, which removes nonlinear transformations and weight matrices entirely. The model retains only linear embedding propagation over the interaction graph, with the final representation obtained by averaging embeddings across layers. This design substantially reduces computational cost while effectively capturing high-order user–item connectivity.

4.7 LightGCL (Graph Contrastive Learning)

LightGCL extends LightGCN by incorporating *Graph Contrastive Learning* to enhance representation robustness. The model generates different *views* of the graph through random perturbations such as edge dropout or embedding noise. A contrastive loss is optimized to pull together representations of the same node across multiple views, yielding a more discriminative latent structure.

4.8 MMGCN

MMGCN (Multi-Modal Graph Convolutional Network) integrates multimodal information (text, images) into the user–item interaction graph to learn semantically enriched representations. By propagating and fusing multimodal features across the graph, MMGCN improves node representations and enhances recommendation accuracy.

4.9 Loss function

The models in this study are trained using the Bayesian Personalized Ranking (BPR) loss function, which is defined as follows:

$$\mathcal{L}_{\mathrm{BPR}} = -\sum_{(u,i,j)} \log \sigma\left(\hat{y}_{ui} - \hat{y}_{uj}\right), \qquad (2)$$

where (u, i, j) denotes a tuple consisting of user u, a positive item i, and a sampled negative item j, and $\sigma(\cdot)$ represents the sigmoid function.

For models incorporating a contrastive learning component, the overall loss function is defined as follows:

$$\mathcal{L} = \mathcal{L}_{\mathrm{BPR}} + \lambda \mathcal{L}_{\mathrm{CL}} + \beta \|\Theta\|_2^2, \qquad (3)$$

where $\mathcal{L}_{\mathrm{CL}}$ is applied only to LightGCL and its variants, $|\Theta|_2^2$ denotes the regularization term to mitigate overfitting, and λ and β are weighting coefficients.

5 Experimental Results

Before training the models, we conducted a preprocessing step to standardize user identifiers in the review dataset. Since the review data does not provide consistent user IDs, we adopted a *cohort-based* approach, in which users are represented as groups of individuals sharing the same demographic characteristics.

Specifically, six attributes—gender (*Gender*), age (*Age*), height (*Height*), weight (*Weight*), shoe size (*Shoe Size*), and country (*Country*)—were combined to form a grouping key. All records sharing the same combination of these attributes were aggregated into a single "user" and assigned a unique integer identifier, which was then used as the user ID for graph construction and model training.

This design enables the application of graph-based recommendation models such as LightGCN and LightGCL under sparse interaction settings. However, it should be noted that the models effectively learn preferences at the *demographic group* level rather than at the individual level. Consequently, evaluation metrics including Precision@K, Recall@K, NDCG@K, and MRR@K reflect recommendation performance at a *segmentation level*. While cohort-based aggregation increases interaction density, it may also dilute preference signals if the defined cohorts are not sufficiently discriminative, which should be considered when interpreting the experimental results.

5.1 Setup

During the experiments, the dataset was split into training, validation, and test sets. In each training batch, *negative sampling* was employed to construct pairwise comparisons for the Bayesian Personalized Ranking (BPR) loss. After training, the models were evaluated using Recall@K, Precision@K, NDCG@K, and MRR@K, which jointly measure both coverage and ranking quality of the recommended lists. During evaluation, items that appeared in the training set for each user were excluded from the recommendation candidates to ensure fairness and to better reflect real-world deployment scenarios.

We compared several graph-based recommendation models built on the user–item interaction graph, including LightGCN, NGCF, LightGCL, and MMGCN. LightGCN was adopted as the baseline model, relying on linear embedding propagation over the interaction graph. NGCF extends LightGCN by incorporating nonlinear transformations and more complex neighborhood aggregation mechanisms. LightGCL further introduces *Graph Contrastive Learning* to enhance the generalization ability of learned embeddings. MMGCN integrates multimodal embeddings to improve recommendation performance.

All models were trained using the BPR loss with *negative sampling*. Text embeddings were extracted using the multilingual-MiniLM-L12-v2 model, while image embeddings were obtained from CLIP ViT-B/32. Evaluation was conducted on the test set using the same strategy of excluding training interactions from the recommendation candidates.

5.2 Results

The experiments conducted in this study include both graph-based recommendation models and traditional recommendation approaches. In addition, we investigate the impact of different similarity measures for traditional models, as well as the effectiveness of various image and text embeddings, and compare their contributions to recommendation performance.

5.2.1 Comparison of Different Models

Table 2 presents the evaluation results of all models on the test set. The metrics Recall@K, Precision@K, NDCG@K, and MRR@K are reported for $K = 5$ and $K = 10$.

As shown in Table 2, LightGCL—and particularly MMGCN—achieves the most robust and superior performance across almost all evaluation metrics (Recall, Precision, NDCG, and MRR) at both Top@5 and Top@10. This indicates their strong ability to effectively combine content information with graph structural signals during representation learning. In contrast, other graph-based models such as NGCF and LightGCN do not demonstrate clear advantages in this setting, which may be attributed to the interaction graph being insufficiently informative or not fully exploited by these architectures.

Interestingly, the content-based approach yields competitive results compared to several purely graph-based models. This can be explained by the fact that content features directly capture user preferences when interaction data is sparse or when the graph structure has not been optimally learned. Overall, the experimental results highlight the critical role of LightGCL and MMGCN, and clearly demonstrate the benefits of integrating content information into graph-based recommendation models, particularly in scenarios where the interaction graph alone is not sufficiently expressive.

5.2.2 Comparison of Different Similarity Measures for Baseline Models

Table 3 reports the evaluation results of different similarity measures applied to the baseline models. For the Mahalanobis similarity, Principal Component Analysis (PCA) was employed to reduce the feature dimensionality to 128, in order to satisfy the assumptions and computational requirements of this metric. The Top@5 results are presented in this table, while the Top@10 results are provided in Appendix A, Table 10.

The results indicate that, for the content-based model, Cosine similarity consistently achieves the best performance across all evaluation metrics (R@5, P@5, N@5, and M@5), demonstrating its suitability for comparing sparse and normalized content representations. Distance-based measures such as Euclidean distance and dot product yield moderate performance, whereas Mahalanobis distance performs significantly worse. This degradation may be attributed to the difficulty of accurately estimating the covariance matrix when dealing with high-dimensional feature spaces and a limited number of samples.

Finally, for the user-based model, overall performance is consistently lower than that of the content-

Model	R@5	P@5	N@5	M@5	R@10	P@10	N@10	M@10
Content-based	0.1053	0.0211	0.0680	0.0557	0.1531	0.0153	0.0834	0.0620
User-based	0.0622	0.0124	0.0465	0.0413	0.0718	0.0072	0.0495	0.0426
NGCF	0.0311	0.0062	0.0170	0.0125	0.0622	0.0062	0.0273	0.0169
LightGCN	0.0287	0.0057	0.0141	0.0095	0.0574	0.0057	0.0232	0.0131
MMGCN	**0.1172**	**0.0234**	**0.0795**	**0.0673**	**0.2057**	**0.0206**	**0.1090**	**0.0800**
LightGCL	**0.1172**	**0.0234**	0.0788	0.0662	**0.1531**	0.0153	0.0901	0.0707

Table 2: Performance comparison of different models in terms of Recall, Precision, NDCG, and MRR. Evaluated at top 5 and top 10.

Model	Similarity	R@5	P@5	N@5	M@5
Content-based	Cosine	**0.1053**	**0.0211**	**0.0680**	**0.0557**
	Mahalanobis	0.0502	0.0100	0.0366	0.0321
	Dot	0.0933	0.0187	0.0613	0.0508
	Euclidean	0.0981	0.0196	0.0601	0.0477
User-based	Cosine	0.0622	0.0124	0.0465	0.0413
	Mahalanobis	0.0072	0.0014	0.0072	0.0072
	Dot	0.0574	0.0115	0.0415	0.0362
	Euclidean	0.0072	0.0014	0.0072	0.0072

Table 3: Comparative analysis of different similarity measures on baseline model performance.

based model across most similarity measures. This observation reflects the inherent challenges of modeling user preferences solely from interaction histories, particularly under sparse data conditions.

5.2.3 Comparison of Different Text Embedding Models

Table 4 presents the evaluation results of different text embedding models when combined with MMGCN and CLIP ViT-B/32 image embeddings. MMGCN is selected for this analysis due to its superior performance compared to other models. The Top@5 results are reported in this table, while the Top@10 results are provided in Appendix A, Table 11.

Overall, Sentence-BERT–based models outperform the other approaches. Among them, *stsb-xlm-r-multilingual* achieves the best results across most evaluation metrics (R@5, P@5, N@5, and M@5), indicating its strong ability to capture semantic similarity in multilingual contexts. The *all-mpnet-base-v2* and *use-cmlm-multilingual* models also deliver competitive performance, particularly in terms of Recall and NDCG, reflecting robust and informative semantic representations.

In contrast, lighter models such as *multilingual-MiniLM-L12-v2* and *multilingual-mpnet-base-v2* yield lower performance, while *LaBSE* shows only moderate effectiveness. This may be attributed to LaBSE being primarily optimized for cross-lingual alignment rather than fine-grained semantic similarity estimation. Overall, these results highlight the importance of selecting appropriate text embedding models to enhance the accuracy of recommendation systems that rely on multilingual textual content.

5.2.4 Comparison of the effectiveness of different image embedding models

Table 5 presents the evaluation results of different image embeddings using the MMGCN model with the multilingual-MiniLM-L12-v2 text embeddings. Metrics at Top@5 are reported in this table, while the corresponding Top@10 results are additionally provided in Appendix A, Table 12.

Overall, models based on CLIP achieve superior performance compared to vision-only models. Among them, CLIP_LAION_B_16 and SigLIP_B16 obtain the highest scores across most evaluation metrics (R@5, P@5, N@5, and M@5), reflecting their ability to learn semantically rich image representations through multimodal pretraining. CLIP_ViT_B_32 yields slightly lower results, while EVA02_CLIP_B16 performs significantly worse, highlighting differences in pretraining strategies and transferability to recommendation tasks. In contrast, vision-only models such as Swin_Tiny, CvT_13, and MaxViT_Tiny consistently achieve

Text Embeddings	R@5	P@5	N@5	M@5
multilingual-MiniLM-L12-v2	0.1268	0.0254	0.0822	0.0676
multilingual-mpnet-base-v2	0.1172	0.0234	0.0771	0.0640
stsb-xlm-r-multilingual	0.1316	**0.0263**	0.0907	**0.0772**
LaBSE	0.1890	0.0189	0.0964	0.0684
use-cmlm-multilingual	**0.2010**	0.0201	0.0987	0.0683
all-mpnet-base-v2	**0.2010**	0.0201	**0.0996**	0.0692

Table 4: Performance comparison of various multilingual text embedding models on recommendation accuracy.

Type	Img Embeddings	R@5	P@5	N@5	M@5
CLIP	CLIP_ViT_B_32	0.1172	0.0234	0.0795	0.0673
	CLIP_LAION_B16	**0.1292**	**0.0258**	0.0812	0.0656
	SigLIP_B16	**0.1292**	**0.0258**	**0.0836**	**0.0689**
	EVA02_CLIP_B16	0.0789	0.0158	0.0475	0.0374
Vision	Swin_Tiny	0.1244	0.0249	0.0773	0.0618
	CvT_13	0.0981	0.0196	0.0659	0.0553
	MaxViT_Tiny	0.1100	0.0220	0.0709	0.0581

Table 5: Performance comparison of various image embedding models on recommendation accuracy.

Model	Text ablation	R@5	P@5	N@5	M@5
Content-based	None	0.1053	0.0211	0.0680	0.0557
	- desc	0.1053	0.0211	0.0680	0.0557
	- mat	0.0933	0.0187	0.0640	0.0544
	- name	0.1053	0.0211	0.0680	0.0557
	- (mat+desc)	0.0933	0.0187	0.0640	0.0544
	- (name+mat)	0.0933	0.0187	0.0640	0.0544
	- (name+desc)	0.1053	0.0211	0.0680	0.0557
MMGCN	None	0.1172	0.0234	0.0795	0.0673
	- desc	**0.1220**	**0.0244**	**0.0823**	**0.0694**
	- mat	0.1100	0.0220	0.0694	0.0562
	- name	**0.1220**	**0.0244**	**0.0823**	**0.0694**
	- (mat+desc)	0.1100	0.0220	0.0694	0.0562
	- (name+mat)	0.1100	0.0220	0.0694	0.0562
	- (name+desc)	**0.1220**	**0.0244**	**0.0823**	**0.0694**
LightGCL	None	0.1172	0.0234	0.0788	0.0662
	- desc	0.1172	0.0234	0.0742	0.0602
	- mat	0.0957	0.0191	0.0697	0.0613
	- name	0.1172	0.0234	0.0742	0.0602
	- (mat+desc)	0.0957	0.0191	0.0697	0.0613
	- (name+mat)	0.0957	0.0191	0.0697	0.0613
	- (name+desc)	0.1172	0.0234	0.0742	0.0602

Table 6: Performance analysis of text ablation on different recommendation models.

lower performance, demonstrating the limitations of relying solely on visual features without semantic alignment with textual information. Overall, these results confirm the importance of multimodal image embedding models—particularly CLIP and SigLIP—in improving the accuracy of image-based recommendation systems.

5.3 Ablation study

In addition to experimenting with different embedding models and similarity measures, we also conducted ablation experiments by removing certain attributes and comparing the effectiveness of the models after ablation. The models considered in this setting are Content-based, MMGCN, and LightGCL, as these are the only models that utilize multimodal attributes.

5.3.1 Text ablation

Table 6 reports the evaluation results from ablation experiments in which different textual attributes are removed. Specifically, desc denotes the product description, mat refers to the material, and name represents the product name. Metrics at Top@5 are presented in this table, while the corresponding Top@10 results are additionally provided in Appendix A, Table 8.

For the Content-based model, removing individual textual components (description, material, or product name) does not lead to significant changes in overall performance, indicating that this model relies relatively evenly on content features and does not strongly exploit the joint use of multiple textual sources. In contrast, for MMGCN, removing either the description or the product name results in a slight increase across all evaluation metrics, while removing the material leads to a slightly smaller impact, suggesting that description and name play key roles in enhancing the model's multimodal representations. A similar trend is observed for LightGCL, where removing the material or material-related combinations causes a noticeable performance degradation, reflecting the model's sensitivity to different sources of content information. Overall, the ablation results do not reveal substantial differences among the effects of removing individual textual features.

5.3.2 Modality ablation

Table 7 presents the evaluation results obtained from ablation experiments in which textual attributes are removed. Here, desc denotes the product description, mat refers to the material, and name represents the product name. Evaluation metrics at Top@5 are reported in this table, while the results at Top@10 are additionally provided in Appendix A, Table 9.

The table compares the performance of different models in ablation experiments that remove textual and visual components, thereby clarifying the contribution of each information modality to recommendation quality. Overall, removing the visual component results in a more pronounced performance degradation than removing the textual component across all three models, particularly in terms of Recall and NDCG, highlighting the important role of visual information in modeling user preferences. For MMGCN, the model maintains relatively strong performance when relying solely on textual information, whereas performance drops substantially when only visual information is used, reflecting its more effective exploitation of linguistic features within the multimodal architecture. A similar trend is observed for LightGCL, where removing textual information leads to a significantly greater performance decline compared to removing visual information, indicating the model's strong dependence on textual features. Overall, the results emphasize that the combination of text and images is essential for achieving optimal performance, while also confirming the central role of textual information in content-integrated graph-based models such as MMGCN and LightGCL.

6 Future work

Although the study achieved positive results, several limitations remain. The dataset is not purely Vietnamese, as product descriptions and fabric technology names often mix Vietnamese and English, posing challenges for monolingual language models. Moreover, the product dataset does not fully cover clothing items that are culturally specific to Vietnam, such as áo dài, which limits the system's generalization ability in broader application contexts. In addition, the scale of interaction data is relatively small and sparse, reducing the effectiveness of purely collaborative filtering methods.

In future work, the project can be extended by leveraging more advanced multimodal embedding models as well as large language models (LLMs) to enhance deep semantic understanding and generate explanations for recommendations. Furthermore, constructing user IDs based on individuals'

Model	Ablation type	R@5	P@5	N@5	M@5
Content-based	Text	0.1053	0.0211	0.0673	0.0549
	Image	0.0646	0.0129	0.0393	0.0311
MMGCN	Text	**0.1124**	**0.0225**	**0.0732**	0.0604
	Image	0.0670	0.0134	0.0418	0.0336
LightGCL	Text	0.0981	0.0196	0.0731	**0.0650**
	Image	0.0263	0.0053	0.0162	0.0128

Table 7: Comparative performance analysis of text-only versus image-only modality ablation.

actual interaction histories rather than cohort-level representations would enable the system to move closer to full personalization. Finally, deploying the system as a real-time application and evaluating it through direct user feedback would be a crucial step toward validating its effectiveness in real-world settings.

7 Conclusion

In this project, we built and evaluated a fashion recommendation system based on real-world data collected from Uniqlo Vietnam, incorporating both product content information and user behavior. By implementing and comparing multiple groups of recommendation approaches simultaneously, the project provides a comprehensive view of the effectiveness and characteristics of each method within the fashion data context. Experimental results demonstrate that graph-based multimodal models, particularly MMGCN and LightGCL, achieve superior performance across most evaluation metrics. The integration of textual and visual information enables the models to better capture product characteristics and user preferences, with textual features playing the most critical role due to their high discriminative power for basic fashion items. In addition, graph contrastive learning mechanisms enhance generalization ability and improve the stability of learned embeddings under conditions of limited interaction data.

References

Xuheng Cai, Chao Huang, Lianghao Xia, and Xubin Ren. 2023. Lightgcl: Simple yet effective graph contrastive learning for recommendation. *Preprint*, arXiv:2302.08191.

Xu Chen, Hanxiong Chen, Hongteng Xu, Yongfeng Zhang, Yixin Cao, Zheng Qin, and Hongyuan Zha. 2019. Personalized fashion recommendation with visual explanations based on multimodal attention network: Towards visually explainable recommendation. In *Proceedings of the 42nd international ACM SIGIR conference on research and development in information retrieval*, pages 765–774.

Dehong Gao, Linbo Jin, Ben Chen, Minghui Qiu, Peng Li, Yi Wei, Yi Hu, and Hao Wang. 2020. Fashionbert: Text and image matching with adaptive loss for cross-modal retrieval. In *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 2251–2260.

Xiangnan He, Kuan Deng, Xiang Wang, Yan Li, Yongdong Zhang, and Meng Wang. 2020. Lightgcn: Simplifying and powering graph convolution network for recommendation. In *Proceedings of the 43rd International ACM SIGIR conference on research and development in Information Retrieval*, pages 639–648.

Xiang Wang, Xiangnan He, Meng Wang, Fuli Feng, and Tat-Seng Chua. 2019. Neural graph collaborative filtering. In *Proceedings of the 42nd international ACM SIGIR conference on Research and development in Information Retrieval*, pages 165–174.

Yinwei Wei, Xiang Wang, Liqiang Nie, Xiangnan He, Richang Hong, and Tat-Seng Chua. 2019. Mmgcn: Multi-modal graph convolution network for personalized recommendation of micro-video. In *Proceedings of the 27th ACM international conference on multimedia*, pages 1437–1445.

A Appendix A: Tables with metrics @10

Due to page size limitations, the @10 metrics are reported in this section.

B Appendix B: Description of attributes in the two datasets

This section includes two tables describing the attributes of the two datasets used in the study.

Model	Text ablation	R@10	P@10	N@10	M@10
Content-based	None	0.1531	0.0153	0.0834	0.0620
	- desc	0.1531	0.0153	0.0834	0.0620
	- mat	0.1603	0.0160	0.0858	0.0634
	- name	0.1531	0.0153	0.0834	0.0620
	- (mat+desc)	0.1603	0.0160	0.0858	0.0634
	- (name+mat)	0.1603	0.0160	0.0858	0.0634
	- (name+desc)	0.1531	0.0153	0.0834	0.0620
MMGCN	None	**0.2057**	**0.0206**	0.1090	0.0800
	- desc	**0.2057**	**0.0206**	**0.1092**	**0.0804**
	- mat	0.2010	0.0201	0.0983	0.0679
	- name	0.2057	**0.0206**	**0.1092**	**0.0804**
	- (mat+desc)	0.2010	0.0201	0.0983	0.0679
	- (name+mat)	0.2010	0.0201	0.0983	0.0679
	- (name+desc) **0.2057**	**0.0206**	**0.1092**	**0.0804**	
LightGCL	None	0.1531	0.0153	0.0901	0.0707
	- desc	0.1699	0.0170	0.0908	0.0668
	- mat	0.1483	0.0148	0.0862	0.0678
	- name	0.1699	0.0170	0.0908	0.0668
	- (mat+desc)	0.1483	0.0148	0.0862	0.0678
	- (name+mat)	0.1483	0.0148	0.0862	0.0678
	- (name+desc)	0.1699	0.0170	0.0908	0.0668

Table 8: Analysis of the performance impact resulting from removing the textual modality across different recommendation models at the @10 metric, presented as a complementary evaluation to Table 6.

Model	Ablation type	R@10	P@10	N@10	M@10
Content-based	Text	0.1651	0.0165	0.0862	0.0624
	Image	0.1244	0.0124	0.0590	0.0394
MMGCN	Text	**0.2010**	**0.0201**	**0.1017**	**0.0721**
	Image	0.1340	0.0134	0.0636	0.0427
LightGCL	Text	0.1435	0.0144	0.0876	0.0709
	Image	0.0526	0.0053	0.0246	0.0162

Table 9: Comparative analysis of model performance between the text-ablation and image-ablation experiments at the @10 metric, provided as a complementary evaluation to the table 7.

Model	Similarity	R@10	P@10	N@10	M@10
Content-based	Cosine	0.1531	0.0153	**0.0834**	**0.0620**
	Mahalanobis	0.0718	0.0072	0.0436	0.0349
	Dot	0.1603	0.0160	0.0827	0.0595
	Euclidean	**0.1722**	**0.0172**	0.0832	0.0568
User-based	Cosine	0.0718	0.0072	0.0495	0.0426
	Mahalanobis	0.0144	0.0014	0.0093	0.0080
	Dot	0.0574	0.0079	0.0483	0.0390
	Euclidean	0.0144	0.0014	0.0093	0.0080

Table 10: Comparative analysis of different similarity measures on the performance of the baseline model at the @10 evaluation metric, provided as a complementary discussion to the table 3.

Text Embeddings	R@10	P@10	N@10	M@10
multilingual-MiniLM-L12-v2	0.2057	0.0206	0.1076	0.0780
multilingual-mpnet-base-v2	**0.2081**	**0.0208**	0.1065	0.0761
stsb-xlm-r-multilingual	0.2057	0.0206	**0.1145**	**0.0870**
LaBSE	0.1890	0.0189	0.0964	0.0684
use-cmlm-multilingual	0.2010	0.0201	0.0987	0.0683
all-mpnet-base-v2	0.2010	0.0201	0.0996	0.0692

Table 11: Comparison of the performance of different multilingual text embedding models on the recommendation accuracy at the @10 metric, provided as a complementary analysis to the table 4.

Type	Img Embeddings	R@10	P@10	N@10	M@10
CLIP	CLIP_ViT_B_32	0.2057	0.0206	0.1090	0.0800
	CLIP_LAION_B16	**0.2129**	**0.0213**	**0.1087**	0.0772
	SigLIP_B16	0.2081	0.0208	0.1085	**0.0788**
	EVA02_CLIP_B16	0.1268	0.0127	0.0632	0.0440
Vision	Swin_Tiny	0.1914	0.0191	0.0984	0.0702
	CvT_13	0.1603	0.0160	0.0861	0.0637
	MaxViT_Tiny	0.1651	0.0165	0.0884	0.0652

Table 12: Comparison of the performance of different image embedding models on the recommendation accuracy at the @10 metric, provided as a complementary analysis to the table 5.

Attribute	Description
category	Product category (e.g., accessories, innerwear, outerwear, etc.).
product_url	URL link to the product detail page on the website.
product_name	Full commercial name of the product.
gender	Target gender of the product (Male/Female/Unisex).
size	Available size variants of the product (One Size, S, M, L, ...).
price	Current selling price (VND).
rate	Average rating score of the product (on a 5.0 scale).
number_of_rates	Total accumulated number of customer ratings.
description	Short textual description of the product (typically includes name and displayed SKU).
material	Detailed information about fabric composition and material ratios.
washing_inst	Manufacturer-provided care and washing instructions.
product_code	Unique identifier (ID) of the product.
img_url	URL to a high-quality representative image of the product.

Table 13: Attributes in the product dataset.

Attribute	Description
product_url	URL of the reviewed product (used to link with the Product table).
product_name	Name of the product at the time of the review.
product_code	Product identifier (foreign key reference).
reviewer	Display name of the customer (with anonymous values "-" removed).
rating	Rating score given by the customer (e.g., "5 out of 5 stars").
comment	Textual content of the detailed review describing the product experience.
review_date	The date on which the customer submitted the review.
gender	Gender of the reviewer.
age	Customer's age group (e.g., 25 to 34 years old).
height	Customer's height range (e.g., 156–160 cm).
weight	Customer's weight range (e.g., 51–55 kg).
shoe_size	Customer's shoe size (if available).
country	Country where the customer resides or made the purchase.

Table 14: Attributes in the reviewer dataset.